Yash Kuletha

Data Scientist | Applied ML Engineer
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Summary

Data science and ML graduate with a published TinyML research paper (anomaly detection, model evaluation), the Stanford/DeepLearning.AI ML Specialization, and 6 months of production experience building data-quality pipelines and AI agent infrastructure for healthcare AI (Python, Ray, Ray Serve, Temporal). Strong in Python and SQL, with exposure to LLM-based and agentic workflows.

Experience

Data Science / AI Infrastructure Intern — XCaliber Health, Bengaluru Jan 2026 – June 2026

- Built Python pipelines for healthcare data ingestion and eCQM processing (scraping, parsing, ETL, validation), cutting processing time by ∼50% and producing clean, validated data for downstream AI use.
- Built distributed workflows with Ray and Daft to transform large-scale structured and unstructured healthcare data, improving throughput by ∼40% over single-node methods.
- Enabled agentic AI deployment with Java control-plane APIs that deployed AI agents as production Ray Serve applications, replacing manual orchestration.

Automation and Analytics Intern — L&T Technology Services, Vadodara May 2024 – July 2024

- Built a Resume Analyzer in Power Apps and Power Automate, automating candidate-screening steps.
- Cleaned recruitment data in Excel and built Power BI dashboards, replacing manual reporting with self-service views.

Projects

TinyML on Microcontrollers — Published Research Paper GitHub

Python, Jupyter, TinyML, Autoencoders, Anomaly Detection, Arduino

- Implemented an autoencoder anomaly detector on Arduino Nano BLE Sense Rev 2 (ECG5000), via a pipeline of architecture search, pruning, quantization-aware training, post-training quantization and benchmarking.
- Beat the baseline paper's approach on recall (94.96% vs 91.07%), F1 (95.54% vs 94.97%) and accuracy (94.82% vs 94.60%); precision was 96.12% vs 99.22%.
- Co-authored "TinyML on Microcontrollers: A Review" (Journal of Advanced Research in Electronics Engineering and Technology, Vol. 13, Issue 1, 2026).

NeoBank — Metadata-Driven Banking Data Pipeline GitHub

Databricks, PySpark, Spark SQL, Delta Lake, Auto Loader, Unity Catalog

- Built a metadata-driven pipeline (table registry, load parameters, watermarks, run audit) ingesting 6 banking datasets from SQL Server (JDBC) and CSV files (Auto Loader), so new tables are onboarded by configuration, not code.
- Implemented FULL, APPEND and Delta MERGE loads with watermark-based incremental processing, logging status, record counts and errors per run, with SMTP run summaries.
- Built 5 Gold tables (customer 360, branch performance, channel summary, daily KPIs, risk summary) served through a dashboard and a Genie agent.

QuickSight Retail — Retail Analytics Lakehouse GitHub

Databricks, PySpark, SQL, Delta Lake, Lakeflow Declarative Pipelines, Unity Catalog

- Integrated Salesforce, PostgreSQL and CSV transactions into a Bronze–Silver–Gold lakehouse, orchestrated by a 5-task Databricks Job.
- Enforced Silver-layer data quality with Lakeflow expectations (validate or drop on IDs, amounts, quantity, discount) and derived fields such as deal-size category and inventory status.
- Modeled a Gold star schema and a Metric View (6 measures, 15+ dimensions) powering a dashboard and Genie interface.

Technical Skills

- **Languages:** Python, SQL, Java, Go
- **ML & Statistics:** Supervised Learning, Classification, Regression, Time Series Forecasting, Statistics, Feature Engineering, Clustering, Recommender Systems, Autoencoders, Anomaly Detection, TinyML
- **Model Evaluation:** Precision/Recall, Cross-Validation, Holdout Testing, Train/Test Splits, Error Analysis
- **Generative AI:** LLMs, Prompt Engineering, LLM-based Workflows, AI Agents, Databricks Genie
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib
- **Data & Platforms:** PySpark, Spark SQL, Ray, Daft, ETL/ELT, Incremental Loading (Delta MERGE), Data Quality Validation, Databricks, Delta Lake, Unity Catalog, Ray Serve, Temporal, REST APIs, Docker
- **Databases & Tools:** PostgreSQL, SQL Server, MySQL, Git, GitHub Actions, Jupyter, Power BI

Education & Certifications

NIIT University — B.Tech, Computer Science (CGPA: 8.51) Aug 2022 – Aug 2026

Machine Learning Specialization — Stanford University / DeepLearning.AI (Coursera) July 2025